

Exam Type: MCQs

Mid Exam

Time Allowed: 20 mins

Name

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Section A: Multiple Choice Questions ($20 \times 1 = 20$ Marks) [CLO-1]

1. Which of the following best describes supervised learning?
a) Learning without labeled data ☐ b) Learning with labeled input-output pairs ✓
c) Grouping data without prior labels ☐ d) Optimizing unsupervised clusters ✓
2. In linear regression, which assumption is not required?
a) Linearity between features and target ☐ b) Homoscedasticity of residuals ✓
c) Independence of observations ☐ d) Features must be categorical ✓
3. The bias-variance trade-off explains:
a) How model interpretability decreases with complexity ☐ b) How underfitting and overfitting affect generalization ✓
c) The relation between model size and runtime ☐ d) How regularization changes optimization speed ✓
4. Which of the following algorithms can model nonlinear relationships?
a) Linear Regression ☐ b) Decision Tree Regressor ✓
c) Logistic Regression ☐ d) Naïve Bayes ✓
5. The mean squared error (MSE) is sensitive to:
a) Outliers ☐ b) Feature scaling ✓
c) Class imbalance ☐ d) Missing data only ✓
6. Logistic regression uses the sigmoid function primarily to:
a) Normalize features ☐ b) Map predictions to probabilities ✓
c) Reduce dimensionality ☐ d) Compute gradients ✓
7. A confusion matrix's off-diagonal elements represent:
a) Correct classifications ☐ b) Incorrect classifications ✓
c) Model accuracy ☐ d) True positives and true negatives ✓
8. The F1-score is most useful when:
a) Data is balanced ☐ b) Precision and recall are equally important ✓
c) True negatives dominate ☐ d) Only recall matters ✓
Precision & Recall
9. K-fold cross-validation helps to:
a) Increase model complexity ☐ b) Reduce training time ✓
c) Estimate model performance stability ☐ d) Directly reduce overfitting ✓
10. The elbow method in clustering is used to:
a) Find optimal number of clusters ✓
b) Reduce dimensionality ☐ c) Improve classification accuracy ☐ d) Calculate cluster centroids ✓
11. In k-means clustering, the objective function minimizes:
a) Sum of distances between centroids ☐ b) Sum of squared distances between points and centroids ✓

c) Within-cluster covariance d) Variance across clusters

12. PCA primarily seeks to:

- a) Maximize data variance in fewer dimensions
- b) Minimize reconstruction error only
- c) Preserve all original features
- d) Optimize classification accuracy

13. In PCA, eigenvalues represent:

- a) The correlation between features
- b) The amount of variance captured by each principal component
- c) The projection direction
- d) The magnitude of feature scaling

14. Which of the following metrics is not appropriate for regression?

- a) Mean Absolute Error
- b) Mean Squared Error
- c) AUC-ROC
- d) Root Mean Squared Error

15. Feature scaling is important because it:

- a) Prevents multicollinearity
- b) Ensures features contribute equally in distance-based models
- c) Always improves accuracy
- d) Reduces the need for validation

16. In decision trees, information gain is computed using:

- a) Mean Squared Error
- b) Gini Index or Entropy
- c) Log-Likelihood
- d) Regularization term

17. The curse of dimensionality refers to:

- a) Exponential growth of data storage
- b) Model accuracy always increasing with more features
- c) Difficulty in distance-based learning in high-dimensional spaces
- d) Inefficiency of PCA in low dimensions

18. Which of the following is a non-parametric model?

- a) Linear Regression
- b) Logistic Regression
- c) K-Nearest Neighbors
- d) Ridge Regression

19. Overfitting is likely to occur when:

- a) Training data is very small and model is simple
- b) Model capacity is too high for the data
- c) Cross-validation is used effectively
- d) Regularization is too strong

20. In a classification model, a high recall but low precision implies:

- a) The model misses many positive cases
- b) The model predicts too many positives, including false ones
- c) The model is underfitting
- d) The model perfectly balances errors

TP
FP
FN
TN

Recall ↑ TPR
Precision ↑ PPV
F1 Score

Exam Type: S/L Answer questions

Mid Exam

Time Allowed: 70 mins

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Section B: Scenario-Based SAQs (3 Scenarios × 3 Qs = 9 Qs, 20 Marks Total) [CLO-3]

Scenario 1: You are training a linear regression model using Gradient Descent. During training, the loss initially decreases but later oscillates (stuck between two values) and fails to converge.

1. What could be the most likely reason for this oscillation?
2. How would you adjust the learning rate or algorithm to fix it? Suggest two changes.
3. Describe how plotting the loss curve over iterations can help diagnose convergence problems in general. What patterns might you look for?

Scenario 2: You trained a simple linear regression model to predict house price from area. After testing, you observe large error values for small houses and smaller error values for large houses.

1. What does this pattern suggest about the model's assumption or fit?
2. Name one diagnostic plot that could help visualize this issue.
3. Suggest one straightforward way to improve the model's predictive performance.

Scenario 3: You apply K-Means clustering on customer transaction data with 20 features and find unstable clusters after each run.

1. Suggest a possible reason for this instability.
2. How could PCA be used before clustering to improve stability?
3. How would you determine the optimal number of clusters objectively?

Section C: Coding-Based Question (2 × 10 = 20 Marks) [CLO-1, CLO-2]

Q1. You are analyzing a dataset `students_performance.csv` containing: `'hours_studied'`, `'attendance'`, `'assignments_completed'`, `'exam_score'`] [CLO-2]

Tasks:

- a) Load and inspect the dataset using pandas, handle missing values.
- b) Split data (80% train, 20% test).
- c) Train a Linear Regression model with scikit-learn to predict `exam_score`.
- d) Compute and report MSE for both training and test sets.
- f) Briefly explain how adding a polynomial term (e.g., `hours_studied2`) might affect bias and variance.

Q2. Write detailed pseudocode for implementing Principal Component Analysis from scratch. [CLO-1]

Then, in 3–5 sentences, discuss:

- Why results depend on the choice of `k`, i.e., no. of principle components?